

S25 — Probability: Addition Rule & Complement

Questions vertical format mein hain. No hints.

1. Addition Rule — Mutually Exclusive Events

A fair die is rolled once.

- Event A = getting **1 or 2**
- Event B = getting **5 or 6**

Answer:

1. Write the sample space.
 2. Are A and B mutually exclusive?
 3. Find .
 4. Find .
 5. Find .
 6. Solve using the addition rule.
 7. Explain why the overlap is zero.
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2. Addition Rule — Overlapping Events

A standard deck of 52 cards is used.

- Event A = selecting a **King**
- Event B = selecting a **Heart**

Answer:

1. Number of Kings.
2. Number of Hearts.

3. Number of cards that are both King and Heart.
 4. Find .
 5. Find .
 6. Find .
 7. Find .
 8. Why can't we simply add ?
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3. Business — Customer Preferences

A survey of 100 customers gives:

Preference	Customers
Mobile App	55
Website	40
Both	20

Answer:

1. What is ?
 2. What is ?
 3. What is ?
 4. Find .
 5. Find the percentage of customers using **neither**.
 6. Why must the overlap be subtracted?
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4. Complement Rule

A company reports that the probability an order is delivered on time is **0.82**.

Answer:

1. What is the probability that an order is **not** delivered on time?
 2. What percentage of orders are expected to be late?
 3. If 500 orders are observed, approximately how many would you expect to be late?
 4. Is the answer in Q3 guaranteed to be exactly that number? Explain.
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5. Identify the Correct Rule

Choose the most appropriate rule:

A. Addition rule

B. Complement rule

1. Probability of a customer buying **Product A or Product B**.
 2. Probability of an order **not being cancelled**, when cancellation probability is known.
 3. Probability of a student being in **Python class or SQL class**.
 4. Probability of a customer **not returning** a product.
 5. Probability of selecting a **King or Queen** from a deck.
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6. Find the Wrong Calculation

An analyst calculates:

They conclude:

Answer:

1. Is the calculation valid?
 2. What is the problem?
 3. What information is missing?
 4. Could the final probability ever be greater than 1?
 5. What would you need to know to correctly calculate ?
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7. Mutually Exclusive vs Overlapping

Classify each pair as **mutually exclusive** or **not mutually exclusive**.

1. Rolling a die: A = even, B = odd.
2. Card: A = King, B = Queen.
3. Card: A = Heart, B = King.
4. Employee: A = Data Analyst, B = Python user.
5. Product order: A = cancelled, B = delivered.

For each one, briefly explain your decision.

8. Business Decision — Marketing Channels

A company tracks 1,000 customers:

- 600 saw an Instagram advertisement.
- 450 saw a YouTube advertisement.
- 250 saw both.

Answer:

1. Find .
 2. Find .
 3. Find .
 4. Find .
 5. Find the number of customers who saw **neither** advertisement.
 6. Which percentage represents customers reached by **at least one** channel?
 7. Why is simply adding $600 + 450$ incorrect?
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9. Complement + Business Interpretation

A fraud-detection system has a **0.06 probability** of flagging an order as fraudulent.

Answer:

1. Probability that an order is **not flagged**.
 2. Percentage of orders not flagged.
 3. Out of 2,000 orders, approximately how many would be flagged?
 4. Out of 2,000 orders, approximately how many would not be flagged?
 5. Does a 6% probability mean exactly 120 orders will always be flagged? Explain.
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10. Full Data Analyst Case

An e-commerce platform analyzes **1,000 customers**.

Customer Behavior	Number
Purchased Product A	500
Purchased Product B	420
Purchased Both A & B	180
Purchased Neither	?

Answer:

1. Calculate .
2. Calculate .
3. Calculate .
4. Calculate .
5. Find the number of customers who purchased **at least one product**.
6. Find the number who purchased **neither product**.
7. Calculate .
8. Are A and B mutually exclusive?
9. Why must the 180 customers who purchased both be subtracted?
10. Give one business interpretation of the result.

11. If the company wants to target customers who purchased neither product, what percentage of the customer base represents this group?